

Cross-Platform Mobile App Development

Week 9 — Mobile Clean Architecture & Testing

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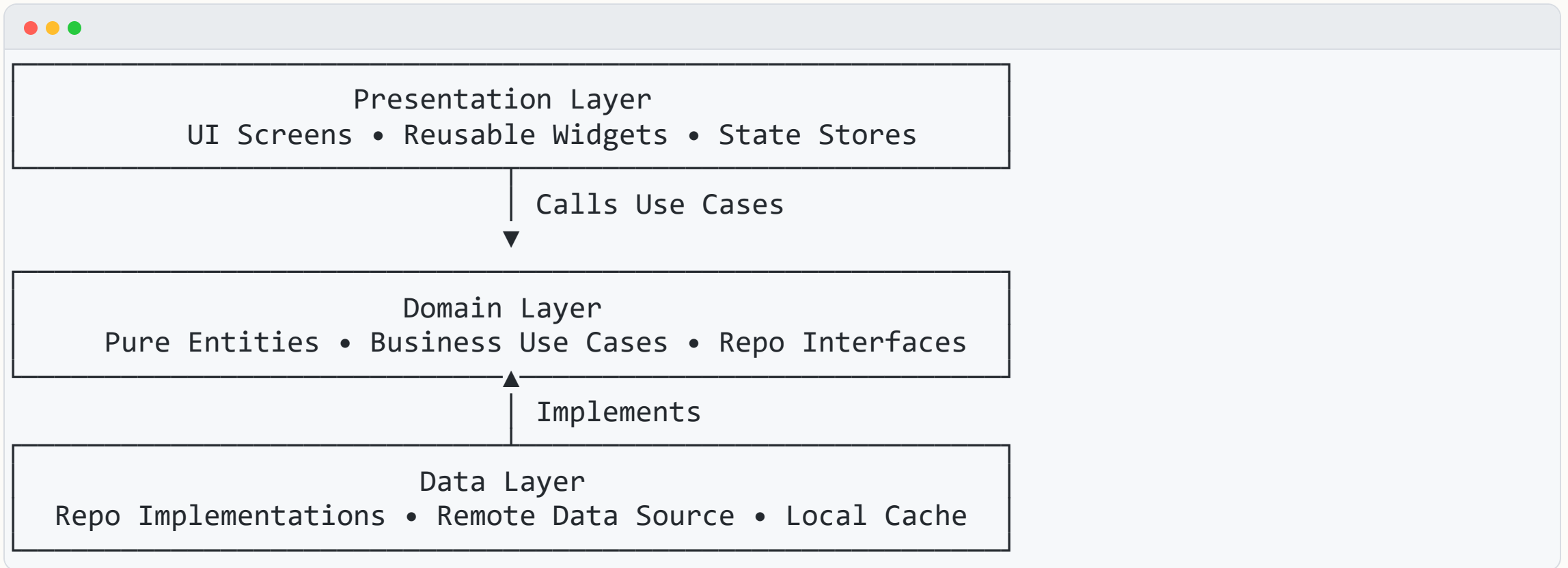
Institution: Faculty of Computer Science, VKU

Today's Agenda

1. The Clean Architecture Philosophy (Robert C. Martin applied to Mobile)
2. Three-Layer Structure: Presentation, Domain, and Data
3. The Repository Pattern: Mediating between Remote API and Local Cache
4. Dependency Injection (DI) & Inversion of Control (IoC) in Mobile
5. Unit Testing pure business logic Use Cases
6. Widget / Component Testing and Mocking Data Sources
7. End-to-End Integration Testing with Maestro / Detox / Patrol

Part 1: Clean Architecture Layers

Three-Layer Architecture



Part 2: Automated Testing Pyramid

Mobile Testing Strategy

-  **Unit Tests (70%):** Fast, deterministic tests for Use Cases, Repositories, and pure utility functions.
-  **Widget / Component Tests (20%):** Verify UI interactions, state updates, and error alerts in isolation.
-  **Integration / E2E Tests (10%):** Automated UI flows running on actual emulators/devices (e.g., full login and booking flow).